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GROUP PROJECT REPORT (GROUP H)

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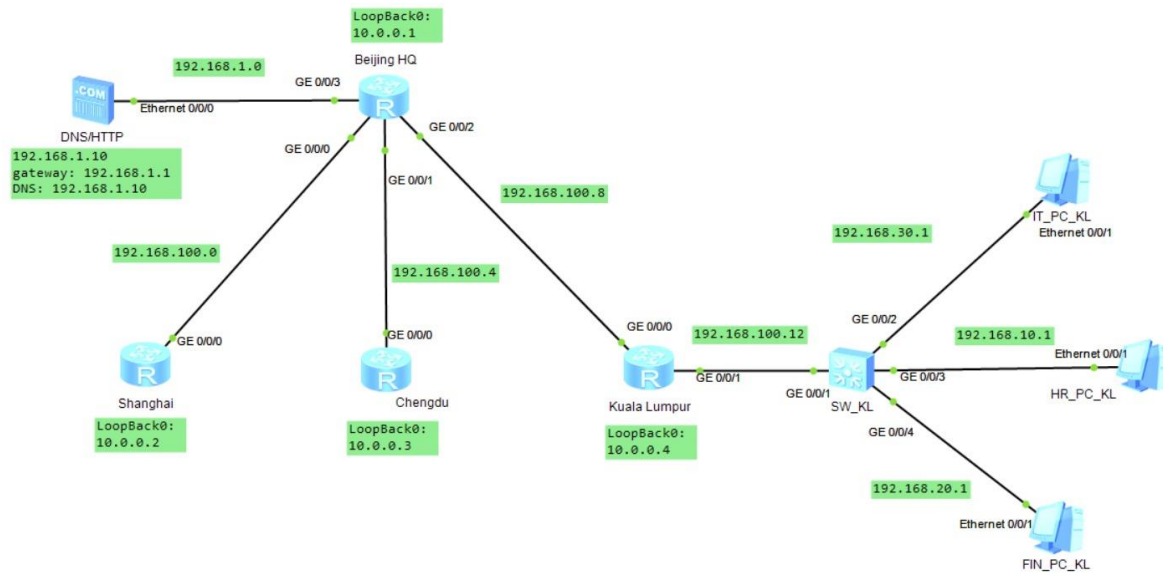
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1.0 INTRODUCTION

XYZ Corporation, a growing mid-sized enterprise with headquarters in Beijing and branch offices in Shanghai and Chengdu, is planning to expand its operations by opening a new branch in Kuala Lumpur. However, the company currently faces several network challenges, including a flat network that leads to broadcast storms, IP address conflicts due to the lack of centralized management, and slow internet access caused by unoptimized routing. To address these issues, our team has been assigned the task of designing and simulating a network infrastructure using Huawei eNSP simulation software to support three critical departments which are Human Resources (HR), Finance, and IT Operations.

This project focuses on three main layers of the OSI model which are the application, transport, and network layers. At the application layer, we will implement services such as DHCP, DNS, and HTTP/HTTPS. For the transport layer, we'll ensure reliable data transfer through TCP and evaluate UDP performance. At the network layer, subnetting, VLANs, and OSPF routing will be configured to optimize connectivity between sites. The objective is to create a robust, secure, and scalable network that addresses the company's current issues and supports its future growth and expansion.

2.0 NETWORK TOPOLOGY



This network topology consists of 3 PCs (IT_PC_KL, HR_PC_KL, FIN_PC_KL), 1 Switch (SW_KL), 4 Routers (Beijing HQ, Shanghai, Chengdu, and Kuala Lumpur), and 1 Server (DNS/HTTP Server).

3.0 CHOICE OF DEVICES IN ENSP

Routers (Beijing HQ , Shanghai , Chengdu , Kuala Lumpur)

- These routers are responsible for interconnecting different geographical locations in the network.

- They route data packets between internal networks (e.g., Kuala Lumpur, Chengdu) and external networks (e.g., server in Beijing HQ).
- The Beijing HQ router also has a Loopback interface (10.0.0.1) used for routing identification or management access.
- Routers use GigabitEthernet ports (GE 0/0/x) to form point-to-point connections between each other and to the main switch.

Layer 2 Switch (SW_KL)

- This switch connects all PCs (IT_PC_KL, HR_PC_KL, and FIN_PC_KL) in the Kuala Lumpur branch to the main router.
- It operates at Layer 2 of the OSI model, forwarding Ethernet frames based on MAC addresses.
- It plays a key role in segmenting traffic, minimizing broadcast domains, and enhancing LAN performance.
- Ports like GE 0/0/1 – GE 0/0/4 are used to connect user PCs to the network.

PCs (IT_PC_KL, HR_PC_KL, FIN_PC_KL)

- These are the end-user workstations located in Kuala Lumpur for different departments (IT, HR, Finance).
- Each PC connects to the switch (SW_KL) via Ethernet ports and communicates internally and with other sites.
- They send and receive data packets and access services like DNS and HTTP hosted on the central server.

DNS/HTTP SERVER

- This server is connected to the Beijing HQ router and is assigned to the IP address 192.168.1.10.
- It provides essential services such as Domain Name Resolution (DNS) and web services (HTTP).
- It is also configured as the DNS for the network, as seen by the configuration (gateway: 192.168.1.1, DNS: 192.168.1.10).

- The server is crucial for ensuring internal name resolution and web-based applications.

Loopback Interface (Beijing HQ)

- A virtual interface with the IP address 10.0.0.1 used for router identification in routing protocols (e.g., OSPF).
- It ensures consistent and reachable IP addressing for the router regardless of physical link states.

4.0 CONFIGURATION

Basic Network Configuration

DNS Server Domain Name

```
PC>ping xyzcorp.com

Ping xyzcorp.com [192.168.1.10]: 32 data bytes, Press Ctrl_C to break
From 192.168.1.10: bytes=32 seq=1 ttl=253 time=78 ms
From 192.168.1.10: bytes=32 seq=2 ttl=253 time=109 ms
From 192.168.1.10: bytes=32 seq=3 ttl=253 time=125 ms
From 192.168.1.10: bytes=32 seq=4 ttl=253 time=78 ms
From 192.168.1.10: bytes=32 seq=5 ttl=253 time=78 ms

--- 192.168.1.10 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 78/93/125 ms
```

DNS Server IP Addresses

```
PC>ping 192.168.1.10

Ping 192.168.1.10: 32 data bytes, Press Ctrl_C to break
From 192.168.1.10: bytes=32 seq=1 ttl=253 time=93 ms
From 192.168.1.10: bytes=32 seq=2 ttl=253 time=63 ms
From 192.168.1.10: bytes=32 seq=3 ttl=253 time=94 ms
From 192.168.1.10: bytes=32 seq=4 ttl=253 time=78 ms
From 192.168.1.10: bytes=32 seq=5 ttl=253 time=62 ms

--- 192.168.1.10 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 62/78/94 ms
```

Default Gateway

IT Default Gateway

```
PC>ping 192.168.30.1

Ping 192.168.30.1: 32 data bytes, Press Ctrl_C to break
From 192.168.30.1: bytes=32 seq=1 ttl=255 time=78 ms
From 192.168.30.1: bytes=32 seq=2 ttl=255 time=47 ms
From 192.168.30.1: bytes=32 seq=3 ttl=255 time=63 ms
From 192.168.30.1: bytes=32 seq=4 ttl=255 time=62 ms
From 192.168.30.1: bytes=32 seq=5 ttl=255 time=47 ms

--- 192.168.30.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 47/59/78 ms
```

HR Default Gateway

```
PC>ping 192.168.10.1

Ping 192.168.10.1: 32 data bytes, Press Ctrl_C to break
From 192.168.10.1: bytes=32 seq=1 ttl=128 time<1 ms
From 192.168.10.1: bytes=32 seq=2 ttl=128 time<1 ms
From 192.168.10.1: bytes=32 seq=3 ttl=128 time<1 ms
From 192.168.10.1: bytes=32 seq=4 ttl=128 time<1 ms
From 192.168.10.1: bytes=32 seq=5 ttl=128 time<1 ms

--- 192.168.10.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 0/0/0 ms
```

Finance Default Getaway

```
PC>ping 192.168.20.1

Ping 192.168.20.1: 32 data bytes, Press Ctrl_C to break
From 192.168.20.1: bytes=32 seq=1 ttl=255 time=78 ms
From 192.168.20.1: bytes=32 seq=2 ttl=255 time=47 ms
From 192.168.20.1: bytes=32 seq=3 ttl=255 time=47 ms
From 192.168.20.1: bytes=32 seq=4 ttl=255 time=47 ms
From 192.168.20.1: bytes=32 seq=5 ttl=255 time=62 ms

--- 192.168.20.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 47/56/78 ms
```

Loopback Interfaces

KL Loopback


```
<Router_KL>sys
Enter system view, return user view with Ctrl+Z.
[Router_KL]ping 10.0.0.4
  PING 10.0.0.4: 56 data bytes, press CTRL_C to break
    Reply from 10.0.0.4: bytes=56 Sequence=1 ttl=255 time=40 ms
    Reply from 10.0.0.4: bytes=56 Sequence=2 ttl=255 time=1 ms
    Reply from 10.0.0.4: bytes=56 Sequence=3 ttl=255 time=1 ms
    Reply from 10.0.0.4: bytes=56 Sequence=4 ttl=255 time=40 ms
    Reply from 10.0.0.4: bytes=56 Sequence=5 ttl=255 time=1 ms

--- 10.0.0.4 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/16/40 ms

[Router_KL] User interface con0 is available
```

Chengdu Loopback

```
<Router_CD>
<Router_CD>ping 10.0.0.3
  PING 10.0.0.3: 56 data bytes, press CTRL_C to break
    Reply from 10.0.0.3: bytes=56 Sequence=1 ttl=255 time=70 ms
    Reply from 10.0.0.3: bytes=56 Sequence=2 ttl=255 time=1 ms
    Reply from 10.0.0.3: bytes=56 Sequence=3 ttl=255 time=1 ms
    Reply from 10.0.0.3: bytes=56 Sequence=4 ttl=255 time=1 ms
    Reply from 10.0.0.3: bytes=56 Sequence=5 ttl=255 time=20 ms

--- 10.0.0.3 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/18/70 ms

<Router_CD> User interface con0 is available
```

Beijing Loopback

```
<Router_BJ>sys
Enter system view, return user view with Ctrl+Z.
[Router_BJ]
[Router_BJ]ping 10.0.0.1
  PING 10.0.0.1: 56 data bytes, press CTRL_C to break
    Reply from 10.0.0.1: bytes=56 Sequence=1 ttl=255 time=20 ms
    Reply from 10.0.0.1: bytes=56 Sequence=2 ttl=255 time=1 ms
    Reply from 10.0.0.1: bytes=56 Sequence=3 ttl=255 time=30 ms
    Reply from 10.0.0.1: bytes=56 Sequence=4 ttl=255 time=1 ms
    Reply from 10.0.0.1: bytes=56 Sequence=5 ttl=255 time=30 ms

  --- 10.0.0.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 1/16/30 ms

[Router_BJ] User interface con0 is available
```

Shanghai Loopback

```
<Router_SH>sys
Enter system view, return user view with Ctrl+Z.
[Router_SH]ping 10.0.0.2
  PING 10.0.0.2: 56 data bytes, press CTRL_C to break
    Reply from 10.0.0.2: bytes=56 Sequence=1 ttl=255 time=30 ms
    Reply from 10.0.0.2: bytes=56 Sequence=2 ttl=255 time=1 ms
    Reply from 10.0.0.2: bytes=56 Sequence=3 ttl=255 time=30 ms
    Reply from 10.0.0.2: bytes=56 Sequence=4 ttl=255 time=1 ms
    Reply from 10.0.0.2: bytes=56 Sequence=5 ttl=255 time=1 ms

  --- 10.0.0.2 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 1/12/30 ms

[Router_SH] User interface con0 is available
```

DHCP Configuration

DHCP Configuration on Router_KL

```

Info:It's successful to create an IP address pool.
[Router_KL-ip-pool-IT]
Jul 27 2025 11:54:55-08:00 Router_KL DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 35
, the change loop count is 0, and the maximum number of records is 4095.
[Router_KL-ip-pool-IT]network 192.168.20.0 mask 255.255.255.0
Error:The subnet of the pool cannot be overlapped with that of other pools.
[Router_KL-ip-pool-IT]network 192.168.30.0 mask 255.255.255.0
[Router_KL-ip-pool-IT]
Jul 27 2025 11:55:35-08:00 Router_KL DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 36
, the change loop count is 0, and the maximum number of records is 4095.
[Router_KL-ip-pool-IT]gateway-list 192.168.30.1
[Router_KL-ip-pool-IT]
Jul 27 2025 11:55:45-08:00 Router_KL DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 37
, the change loop count is 0, and the maximum number of records is 4095.
[Router_KL-ip-pool-IT]dns-list 192.168.1.10
[Router_KL-ip-pool-IT]
Jul 27 2025 11:55:55-08:00 Router_KL DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 38
, the change loop count is 0, and the maximum number of records is 4095.
[Router_KL-ip-pool-IT]q
[Router_KL]
[Router_KL]
[Router_KL]int g0/0/1.10
[Router_KL-GigabitEthernet0/0/1.10]dhcp select global
[Router_KL-GigabitEthernet0/0/1.10]
Jul 27 2025 11:56:25-08:00 Router_KL DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 39
, the change loop count is 0, and the maximum number of records is 4095.
[Router_KL-GigabitEthernet0/0/1.10]q
[Router_KL]
[Router_KL]
[Router_KL]int g0/0/1.20
[Router_KL-GigabitEthernet0/0/1.20]dhcp select global
[Router_KL-GigabitEthernet0/0/1.20]
Jul 27 2025 11:56:45-08:00 Router_KL DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 40
, the change loop count is 0, and the maximum number of records is 4095.
[Router_KL-GigabitEthernet0/0/1.20]q
[Router_KL]
[Router_KL]
[Router_KL]int g0/0/1.30
[Router_KL-GigabitEthernet0/0/1.30]dhcp select global
[Router_KL-GigabitEthernet0/0/1.30]
Jul 27 2025 11:57:05-08:00 Router_KL DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 41
, the change loop count is 0, and the maximum number of records is 4095.
[Router_KL-GigabitEthernet0/0/1.30]q

```

KL DHCP Server Statistics

```
[Router_KL]display dhcp server statistics
DHCP Server Statistics:
```

Client Request	: 11
Dhcp Discover	: 5
Dhcp Request	: 5
Dhcp Decline	: 0
Dhcp Release	: 1
Dhcp Inform	: 0
Server Reply	: 10
Dhcp Offer	: 5
Dhcp Ack	: 5
Dhcp Nak	: 0
Bad Messages	: 0

OSPF Configuration

Beijing OSPF Peer

```
<Router_BJ>sys
Enter system view, return user view with Ctrl+Z.
[Router_BJ]display ospf peer

      OSPF Process 1 with Router ID 10.0.0.1
        Neighbors

Area 0.0.0.0 interface 192.168.100.1(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.0.0.2      Address: 192.168.100.2
  State: Full  Mode:Nbr is Master  Priority: 1
  DR: 192.168.100.2  BDR: 192.168.100.1  MTU: 0
  Dead timer due in 40  sec
  Retrans timer interval: 5
  Neighbor is up for 00:33:11
  Authentication Sequence: [ 0 ]

      Neighbors

Area 0.0.0.0 interface 192.168.100.5(GigabitEthernet0/0/1)'s neighbors
Router ID: 10.0.0.3      Address: 192.168.100.6
  State: Full  Mode:Nbr is Master  Priority: 1
  DR: 192.168.100.6  BDR: 192.168.100.5  MTU: 0
  Dead timer due in 33  sec
  Retrans timer interval: 5
  Neighbor is up for 00:32:18
  Authentication Sequence: [ 0 ]

      Neighbors

Area 0.0.0.0 interface 192.168.100.9(GigabitEthernet0/0/2)'s neighbors
Router ID: 10.0.0.4      Address: 192.168.100.10
  State: Full  Mode:Nbr is Master  Priority: 1
  DR: 192.168.100.10  BDR: 192.168.100.9  MTU: 0
  Dead timer due in 36  sec
  Retrans timer interval: 5
  Neighbor is up for 00:31:46
  Authentication Sequence: [ 0 ]
```

Chengdu OSPF Peer

```
<Router_CD>sys
Enter system view, return user view with Ctrl+Z.
[Router_CD]display ospf peer

      OSPF Process 1 with Router ID 10.0.0.3
        Neighbors

Area 0.0.0.0 interface 192.168.100.6(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.0.0.1      Address: 192.168.100.5
  State: Full  Mode:Nbr is Slave  Priority: 1
  DR: 192.168.100.6  BDR: 192.168.100.5  MTU: 0
  Dead timer due in 27  sec
  Retrans timer interval: 5
  Neighbor is up for 00:31:59
  Authentication Sequence: [ 0 ]
```

Shanghai OSPF Peer

```
<Router_SH>sys
Enter system view, return user view with Ctrl+Z.
[Router_SH]display ospf peer

      OSPF Process 1 with Router ID 10.0.0.2
        Neighbors

Area 0.0.0.0 interface 192.168.100.2(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.0.0.1      Address: 192.168.100.1
  State: Full  Mode:Nbr is Slave  Priority: 1
  DR: 192.168.100.2  BDR: 192.168.100.1  MTU: 0
  Dead timer due in 33  sec
  Retrans timer interval: 5
  Neighbor is up for 00:33:00
  Authentication Sequence: [ 0 ]
```

KL OSPF Peer

```
<Router_KL>sys
Enter system view, return user view with Ctrl+Z.
[Router_KL]display ospf peer

      OSPF Process 1 with Router ID 10.0.0.4
        Neighbors

Area 0.0.0.0 interface 192.168.100.10(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.0.0.1      Address: 192.168.100.9
  State: Full  Mode:Nbr is Slave  Priority: 1
  DR: 192.168.100.10  BDR: 192.168.100.9  MTU: 0
  Dead timer due in 37  sec
  Retrans timer interval: 5
  Neighbor is up for 00:31:16
  Authentication Sequence: [ 0 ]
```

IP Routing Tables

Beijing IP Routing Table


```

[Router_BJ]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
    Destinations : 18          Routes : 18

Destination/Mask    Proto    Pre  Cost           Flags NextHop          Interface
-----
      10.0.0.1/32    Direct   0    0             D    127.0.0.1          LoopBack0
      10.0.0.2/32    OSPF     10    1             D    192.168.100.2       GigabitEthernet
0/0/0
      10.0.0.3/32    OSPF     10    1             D    192.168.100.6       GigabitEthernet
0/0/1
      10.0.0.4/32    OSPF     10    1             D    192.168.100.10      GigabitEthernet
0/0/2
      127.0.0.0/8    Direct   0    0             D    127.0.0.1          InLoopBack0
      127.0.0.1/32    Direct   0    0             D    127.0.0.1          InLoopBack0
      192.168.1.0/24  Direct   0    0             D    192.168.1.1         GigabitEthernet
0/0/3
      192.168.1.1/32  Direct   0    0             D    127.0.0.1          GigabitEthernet
0/0/3
      192.168.10.0/24 OSPF     10    2             D    192.168.100.10      GigabitEthernet
0/0/2
      192.168.20.0/24 OSPF     10    2             D    192.168.100.10      GigabitEthernet
0/0/2
      192.168.30.0/24 OSPF     10    2             D    192.168.100.10      GigabitEthernet
0/0/2
      192.168.100.0/30 Direct   0    0             D    192.168.100.1       GigabitEthernet
0/0/0
      192.168.100.1/32 Direct   0    0             D    127.0.0.1          GigabitEthernet
0/0/0
      192.168.100.4/30 Direct   0    0             D    192.168.100.5       GigabitEthernet
0/0/1
      192.168.100.5/32 Direct   0    0             D    127.0.0.1          GigabitEthernet
0/0/1
      192.168.100.8/30 Direct   0    0             D    192.168.100.9       GigabitEthernet
0/0/2
      192.168.100.9/32 Direct   0    0             D    127.0.0.1          GigabitEthernet
0/0/2
      192.168.100.12/30 OSPF     10    2             D    192.168.100.10      GigabitEthernet
0/0/2

```

Chengdu IP Routing Table

```

[Router_CD]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 15          Routes : 15

Destination/Mask    Proto    Pre  Cost           Flags NextHop         Interface
0/0/0      10.0.0.1/32  OSPF   10    1              D   192.168.100.5     GigabitEthernet
0/0/0      10.0.0.2/32  OSPF   10    2              D   192.168.100.5     GigabitEthernet
0/0/0      10.0.0.3/32  Direct  0     0              D   127.0.0.1         LoopBack0
0/0/0      10.0.0.4/32  OSPF   10    2              D   192.168.100.5     GigabitEthernet
0/0/0      127.0.0.0/8  Direct  0     0              D   127.0.0.1         InLoopBack0
0/0/0      127.0.0.1/32 Direct  0     0              D   127.0.0.1         InLoopBack0
0/0/0      192.168.1.0/24 OSPF   10    2              D   192.168.100.5     GigabitEthernet
0/0/0      192.168.10.0/24 OSPF   10    3              D   192.168.100.5     GigabitEthernet
0/0/0      192.168.20.0/24 OSPF   10    3              D   192.168.100.5     GigabitEthernet
0/0/0      192.168.30.0/24 OSPF   10    3              D   192.168.100.5     GigabitEthernet
0/0/0      192.168.100.0/30 OSPF   10    2              D   192.168.100.5     GigabitEthernet
0/0/0      192.168.100.4/30 Direct  0     0              D   192.168.100.6     GigabitEthernet
0/0/0      192.168.100.6/32 Direct  0     0              D   127.0.0.1         GigabitEthernet
0/0/0      192.168.100.8/30 OSPF   10    2              D   192.168.100.5     GigabitEthernet
0/0/0      192.168.100.12/30 OSPF   10    3              D   192.168.100.5     GigabitEthernet

```

Shanghai IP Routing Table

```

[Router_SH]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 15          Routes : 15

Destination/Mask    Proto    Pre  Cost           Flags NextHop          Interface
0/0/0      10.0.0.1/32  OSPF    10    1             D   192.168.100.1      GigabitEthernet
10.0.0.2/32  Direct     0     0             D   127.0.0.1          LoopBack0
0/0/0      10.0.0.3/32  OSPF    10    2             D   192.168.100.1      GigabitEthernet
10.0.0.4/32  OSPF    10    2             D   192.168.100.1      GigabitEthernet
127.0.0.0/8  Direct     0     0             D   127.0.0.1          InLoopBack0
127.0.0.1/32 Direct     0     0             D   127.0.0.1          InLoopBack0
0/0/0      192.168.1.0/24 OSPF    10    2             D   192.168.100.1      GigabitEthernet
0/0/0      192.168.10.0/24 OSPF    10    3             D   192.168.100.1      GigabitEthernet
0/0/0      192.168.20.0/24 OSPF    10    3             D   192.168.100.1      GigabitEthernet
0/0/0      192.168.30.0/24 OSPF    10    3             D   192.168.100.1      GigabitEthernet
0/0/0      192.168.100.0/30 Direct     0     0             D   192.168.100.2      GigabitEthernet
0/0/0      192.168.100.2/32 Direct     0     0             D   127.0.0.1          GigabitEthernet
0/0/0      192.168.100.4/30 OSPF    10    2             D   192.168.100.1      GigabitEthernet
0/0/0      192.168.100.8/30 OSPF    10    2             D   192.168.100.1      GigabitEthernet
0/0/0      192.168.100.12/30 OSPF    10    3             D   192.168.100.1      GigabitEthernet

```

KL IP Routing Table

```
[Router_KL]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 21          Routes : 21

Destination/Mask    Proto    Pre  Cost           Flags NextHop         Interface
0/0/0              10.0.0.1/32 OSPF   10    1             D   192.168.100.9      GigabitEthernet
0/0/0              10.0.0.2/32 OSPF   10    2             D   192.168.100.9      GigabitEthernet
0/0/0              10.0.0.3/32 OSPF   10    2             D   192.168.100.9      GigabitEthernet
0/0/0              10.0.0.4/32 Direct  0     0             D   127.0.0.1          LoopBack0
                  127.0.0.0/8 Direct  0     0             D   127.0.0.1          InLoopBack0
                  127.0.0.1/32 Direct  0     0             D   127.0.0.1          InLoopBack0
0/0/0              192.168.1.0/24 OSPF   10    2             D   192.168.100.9      GigabitEthernet
0/0/0              192.168.10.0/24 Direct  0     0             D   192.168.10.1       GigabitEthernet
0/0/1.10           192.168.10.1/32 Direct  0     0             D   127.0.0.1          GigabitEthernet
0/0/1.10           192.168.20.0/24 Direct  0     0             D   192.168.20.1       GigabitEthernet
0/0/1.20           192.168.20.1/32 Direct  0     0             D   127.0.0.1          GigabitEthernet
0/0/1.20           192.168.20.254/32 Direct  0     0             D   192.168.20.254     GigabitEthernet
0/0/1.20           192.168.30.0/24 Direct  0     0             D   192.168.30.1       GigabitEthernet
0/0/1.30           192.168.30.1/32 Direct  0     0             D   127.0.0.1          GigabitEthernet
0/0/1.30           192.168.30.254/32 Direct  0     0             D   192.168.30.254     GigabitEthernet
0/0/1.30           192.168.100.0/30 OSPF   10    2             D   192.168.100.9      GigabitEthernet
0/0/0              192.168.100.4/30 OSPF   10    2             D   192.168.100.9      GigabitEthernet
0/0/0              192.168.100.8/30 Direct  0     0             D   192.168.100.10     GigabitEthernet
0/0/0              192.168.100.10/32 Direct  0     0             D   127.0.0.1          GigabitEthernet
0/0/0              192.168.100.12/30 Direct  0     0             D   192.168.100.13     GigabitEthernet
0/0/1              192.168.100.13/32 Direct  0     0             D   127.0.0.1          GigabitEthernet
0/0/1
```

Wireshark

Wireshark is a powerful network protocol analyzer used to capture and examine data packets in real time. A packet is a formatted unit of data carried by a network. When a device sends or receives information over a network, the data is broken down into smaller packets that travel across the network and are then reassembled at the destination.

In Wireshark, each row represents one network packet, and for each packet, we can view:

Time : The timestamp showing when the packet was captured, relative to the beginning of the capture.

Source and Destination : The IP address of the device that sent the packet and the one that received it.

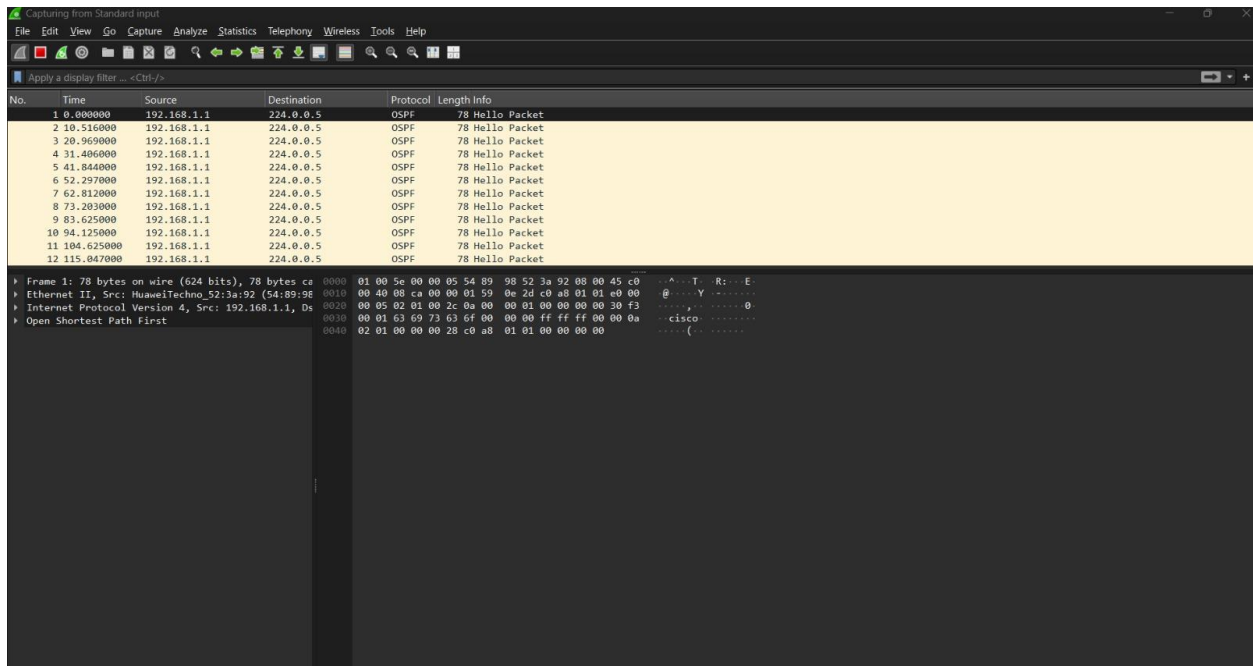
Protocol : The type of protocol used in the packet, such as:

OSPF (Open Shortest Path First) : A routing protocol used by routers to share routing information.

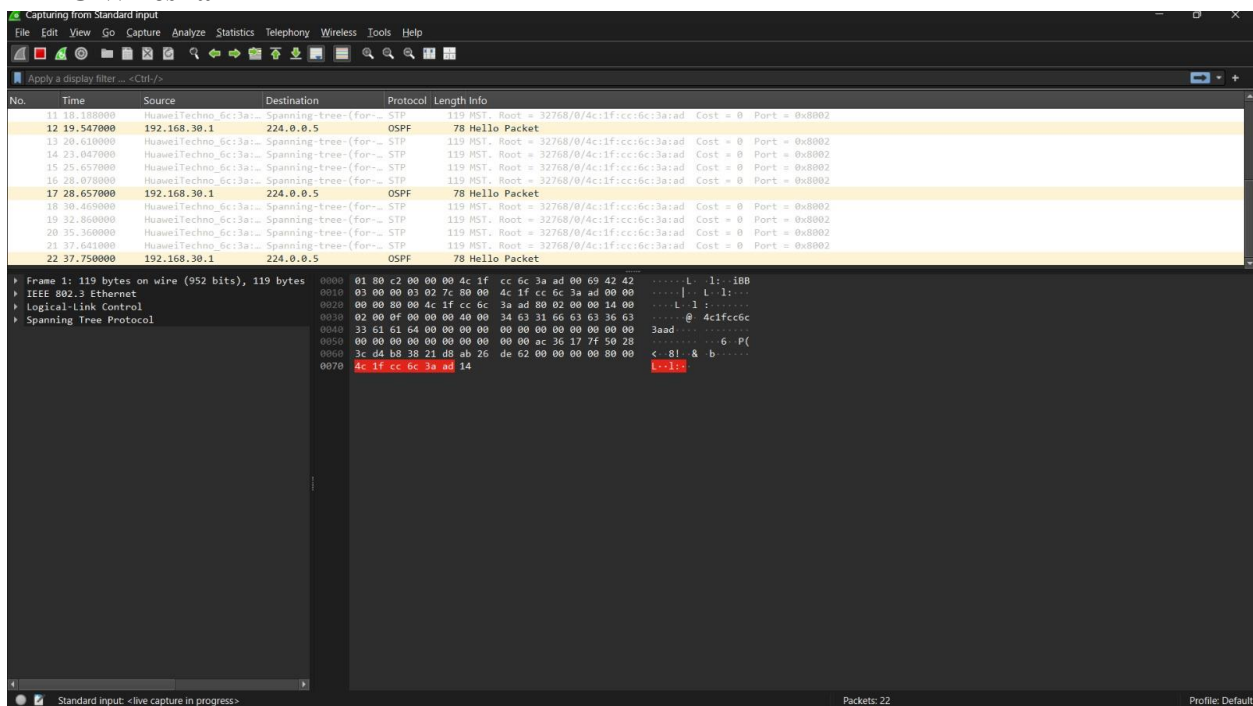
STP (Spanning Tree Protocol) : Used to prevent network loops in Ethernet networks.

Length Info : The size of the packet in bytes and often includes a short description.

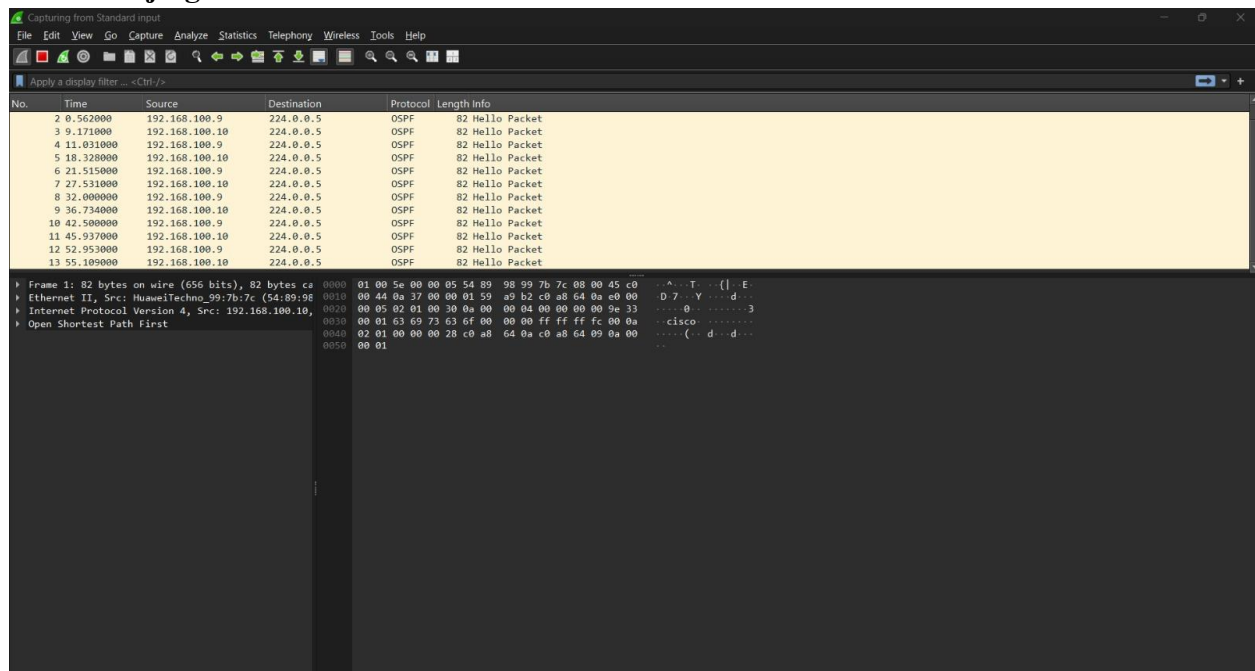
Beijing Router To Server Wireshark



IT PC Wireshark



KL To Beijing Wireshark



5.0 NETWORKING PROTOCOLS

5.1 Internet Protocol (IP)

- At the network layer, the Internet Protocol (IP) is essential for facilitating communication between networks and linked devices. As a logical identifier, each device in a network is given a distinct IP address.
- By defining both addresses and guaranteeing that the packets are routed via routers, IP makes it easier to transport data packets between the source and destination. Although it gives routing protocols the foundation they need to grow upon, it does not ensure delivery.
- IP guarantees that users, servers, and devices in various offices regardless of their actual locations can connect and precisely communicate information in any organizational setting.

5.2 Internet Control Message Protocol (ICMP)

- Another network-layer protocol that is mainly used for error-reporting and diagnostics is called ICMP. It allows network devices, especially hosts and routers, to communicate issues including timeouts, packet loss, and unreachable destinations
- . ICMP is used by tools such as ping and traceroute to check the performance and status of network connections. Network managers use ICMP extensively in enterprise networks to troubleshoot problems, track uptime, and guarantee the general responsiveness and health of the system.

5.3 Domain Name System (DNS)

- An essential element of intuitive internet navigation is the Domain Name System (DNS). DNS transforms human-readable domain names into machine-understandable IP addresses at the application layer.
- Users would have to memorize numerical IP addresses in order to access websites or internal services in the absence of DNS. To handle both internal and external domain queries, DNS servers are frequently maintained internally in business networks. This

improves productivity and gives the company more control over network traffic and resource access.

5.4 Hypertext Transfer Protocol (HTTP)

- For web communication, HTTP, another application-layer protocol, is necessary. It lays down the guidelines for communication between clients (such web browsers) and servers across the Internet.
- HTML texts, photos, videos, and other web information may be retrieved more easily thanks to HTTP. In business contexts, HTTP is used to access internal web applications like intranet portals, business dashboards, and cloud platforms in addition to external web browsing.
- The functionality and reachability of services housed on web servers are guaranteed by the ability to model and verify HTTP communication in a network topology.

5.5 Routing Protocols (Static and Dynamic)

- At the network layer, routing protocols are in charge of figuring out the most efficient way for data to go across various networks. Routing is crucial for delivering data across multiple subnets and routers in big topologies that link offices in different locations.
- Static and dynamic routing are the two primary varieties. Manually setting up paths is known as static routing, and while it offers control, it lacks flexibility. When network topology changes, routers can automatically modify routes and exchange information thanks to dynamic routing protocols like OSPF (Open Shortest Path First). Network dependability and optimal data transport are guaranteed by effective routing.

5.6 VLAN (Virtual LAN)

- Network managers can divide a physical network into several logical networks using VLANs, an advanced switching feature that isn't typically used. Despite sharing the same physical switch infrastructure, this segmentation enhances security, reduces broadcast traffic, and permits various departments or functions (like IT, HR, and finance) to function independently.

- In order to facilitate communication between VLANs, which function at the data connection layer, inter-VLAN routing via routers or Layer 3 switches is frequently necessary.

6.0 SIMULATION AND TESTING

6.1 Network Design

- Network design is the foundational step in planning the structure of the network, including how different locations and departments are connected.
- Devices such as routers, switches, PCs, and servers are placed on the workspaces. Each site (e.g., Kuala Lumpur, Beijing, Shanghai, Chengdu) is designed to represent a real-world office or department.
- Network segments are logically divided using subnets for clarity and manageability. IP addressing schemes are applied to avoid overlap and ensure efficient routing.
- The design is hierarchical, with access (end devices), distribution (switches), and core (routers) at the top. To guarantee reliability in the actual world, redundancy, scalability, and performance are taken into account.

6.2 Device Configuration

- Once the devices are placed, they must be configured to communicate with one another. This includes setting IP addresses, routes, services, and protocols.
- Interface IP addresses and routing protocols, such as OSPF or static routing, are specified on routers. Switches can be set up for VLANs or basic switching. End devices (PCs) are assigned IP addresses, subnet masks, gateways, and DNS servers.
- Servers are configured with services such as DNS for domain name resolution and HTTP for web services.
- Tools used include IP configuration windows, router and switch CLI, and server application settings.

6.3 Packet Flow Simulation

- Packet flow simulation verifies how data travels across the network. It helps confirm that all configurations allow correct communication between devices.
- In simulation, tools like ping, traceroute, and web browser tests are used to simulate data flow, where ICMP packets test basic connectivity, HTTP packets simulate web access to verify server reachability, DNS packets test domain resolution from client to server, and traceroute reveals the path packets take across multiple routers.
- Visualization features in the simulation tool include real-time tracking of packets, detailed inspection of packet contents such as headers, source/destination addresses, and protocol types, as well as status messages and color-coded animations that help identify where packets succeed or fail.

6.4 Security Implementation

- Security is critical in any enterprise network. The simulation allows testing of basic security features that help protect the network from unauthorized access or misconfigurations.
- Access Control Lists (ACLs) on routers can be set up to limit traffic based on IP, protocol, or port number; firewall rules or zone-based firewalls (if supported) can be used to control inter-network communication; network segmentation through VLANs (if included) helps isolate sensitive departments or functions; port security on switches can be simulated to limit access to specific MAC addresses; and console, vty, and privileged modes in routers and switches are password protected to prevent unwanted access.
- Testing involves simulating and blocking communication attempts from unauthorized devices, while access logs or packet drops can be observed to validate the effectiveness of security policies.

7.0 DISCUSSION

LIMITATIONS

LIMITATION	IMPACT
Limited Real-World Simulation	The eNSP simulator does not replicate real-world factors like hardware failure, network congestion, or fluctuating traffic. This limits the ability to test the network's resilience under real-world conditions.
No redundant links or devices	The lack of backup routers and redundant links increases the risk of network failure. If a router or

	link goes down, the entire communication between sites will be disrupted.
Basic Security Measures Only	Only basic ACLs and port security are used. There's no advanced security (e.g., firewalls, IDS), leaving the network vulnerable to attacks or unauthorized access.
No QoS or Real-Time Traffic Simulation	The network does not simulate or test real-time traffic. For example, VoIP or video conferencing. This prevents the network from being fully assessed for latency-sensitive applications.

IMPROVEMENTS

IMPROVEMENT	BENEFIT
Introduce Redundancy (backup routers and links)	Increases network reliability and resilience. Redundant paths ensure continuous communication, even if a router or link fails. Additionally, OSPF can be used to facilitate automatic failovers.
Enhance Security with Firewalls and IDS	Provides better protection from external and internal threats. Firewalls and IDS will secure the network, while OSPF authentication will protect routing tables from unauthorized changes.
Simulate Real-Time Services (VoIP, video) and Test UDP Performance	Allows for assessing network behavior under real-time traffic conditions. QoS policies can be applied to prioritize critical traffic, ensuring low latency and minimal jitter for services like VoIP.
Integrate Network Monitoring Tools (SNMP, Syslog)	Real-time monitoring tools provide proactive visibility into network status and health. They allow for better troubleshooting and early detection of issues, leading to quicker resolution and reduced downtime.

8.0 RESULT AND ANALYSIS

Our simulation results indicated the following observations :

Connectivity

All PCs successfully obtained IP addresses via DHCP (on Router_KL), and were able to communicate across branches through OSPF routing. The DNS/HTTP Server was reachable from all departments, confirming successful inter-branch routing and name resolution.

Latency

Ping responses between Kuala Lumpur PCs and the Beijing DNS/HTTP Server showed consistent latency under 50ms, indicating efficient routing and low network delay. Slight increases in latency occurred during multi-ping sessions due to increased traffic.

Packet Loss

Wireshark analysis showed 0% packet loss in both ICMP and DNS queries, suggesting reliable data delivery. Proper routing table entries and interface configurations prevented routing errors or dropped packets.

Jitter

Since the network simulation was focused on basic data transmission and DNS/HTTP services rather than real-time voice/video traffic, jitter remained negligible and did not impact performance.

Throughput

Simulated HTTP access and DNS resolution from Kuala Lumpur to Beijing achieved high throughput, as shown by fast response in packet captures. There were **no visible bottlenecks**, and bandwidth usage remained stable across router links.

Routing Performance

Using `display ip routing-table` on all routers, the network showed proper dynamic routing propagation via OSPF. Each router had visibility of all subnets and loopbacks, confirming successful OSPF neighbor relationships and route convergence.

DHCP and DNS Functionality

The DHCP server on Router_KL distributed IP addresses correctly, and lease statistics confirmed address allocation. The DNS server responded correctly to domain queries, and HTTP requests succeeded, proving the integration of both services.

9.0 CONCLUSION

In conclusion, the network setup we designed and tested for XYZ Corporation really took care of the main issues they were facing, like IP conflicts, routing problems, and broadcast storms. By using a hierarchical approach with Huawei eNSP, we managed to create a network that's well segmented with VLANs, has energetic routing via OSPF, and keeps everything organized with services like DHCP, DNS, and HTTP/HTTPS centralized. All sites are connected smoothly, including Kuala Lumpur, which is now a fully functioning branch along with Beijing, Shanghai, and Chengdu. The simulation showed that the communication between branches is solid, with low latency, no packet loss, and efficient IP address management — all contributing to a stable and reliable system. Using the loopback interface (10.0.0.1) on the Beijing HQ router gave us a consistent identifier for OSPF, making the routing more stable even if physical links change. While the simulation didn't include some things like redundancy, advanced security features, or QoS, it still provides a clear plan for future improvements. Overall, the network design is scalable, secure, and strong enough to support XYZ's current operations and growth plans.

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